CO-TRIMOXAZOLE

Hyperkalaemia
Severe skin reactions

CHEMICAL NAME Trimethoprim: 5-(3,4,5-trimethoxybenzyl)pyrimidine-2,4-diamine + sulfamethoxazole: 4-amino-N-(5-methyl-1,2-oxazol-3-yl)benzenesulfonamide (1:5)

CLASS Sulfonamide + dihydrofolate reductase inhibitor combination

TRADE NAMES Bactrim · Septrin · Septra · Trimeth

THERAPEUTIC USES / INDICATIONS

- *Pneumocystis jirovecii* pneumonia; treatment and prophylaxis (HIV, transplant, chemotherapy).
- Urinary tract infections; prostatitis.
- Toxoplasmosis prophylaxis; nocardiosis; *Stenotrophomonas*.
- MRSA skin infections (community); brucellosis (children).

ADMINISTRATION

- **PO** tablets (480 mg, 960 mg DS) or suspension with plenty of fluid.
- **IV** infusion diluted, over 60–90 min.
- Maintain fluid intake ≥1.5 L/day (prevent crystalluria).

DOSAGES

- **Adult UTI:** 960 mg every 12 h.
- **PCP treatment:** 120 mg/kg/day in 2–4 divided doses for 14–21 days.
- **PCP prophylaxis:** 960 mg once daily or 3 times weekly (480 mg daily also used).
- **Children:** 24 mg/kg every 12 h (trimethoprim 4 mg/kg).
- CrCl 15–30: half dose; avoid <15.

PHARMACOKINETICS

- **Peak:** 1–4 h. Well absorbed.
- Good tissue penetration incl. CSF, prostate.
- Metabolised in liver; excreted in urine. **t½:** TMP 10 h, SMX 10 h.

PHARMACODYNAMICS

- Sequential blockade of **folate synthesis**: sulfamethoxazole inhibits dihydropteroate synthase; trimethoprim inhibits dihydrofolate reductase → bactericidal.
- Trimethoprim blocks ENaC in distal tubule (amiloride-like) → hyperkalaemia.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, anorexia.
- Rash (common, especially in HIV).
- Glossitis.

ADVERSE EFFECTS (SERIOUS)

- SJS/TEN; DRESS.
- Hyperkalaemia; raised creatinine.
- Bone-marrow suppression (folate deficiency); haemolysis in G6PD deficiency.
- Hepatitis; aseptic meningitis; hypoglycaemia.

CONTRAINDICATIONS & CAUTIONS

- Sulfonamide or trimethoprim hypersensitivity.
- Severe hepatic or renal impairment; acute porphyria.
- Infants <6 weeks (kernicterus); late pregnancy.
- **Caution:** G6PD deficiency, folate deficiency, elderly, first trimester.

KEY INTERACTIONS

- **Methotrexate:** ↑↑ toxicity (pancytopenia).
- **Warfarin:** ↑ INR; **phenytoin:** ↑ levels.
- **ACE inhibitors, ARBs, spironolactone:** hyperkalaemia.
- **Sulfonylureas:** hypoglycaemia.

NURSING MANAGEMENT

- Check sulfa allergy and G6PD status.
- Encourage fluids; monitor urine output.
- Monitor K^+, creatinine, CBC in high doses or long-term.
- Stop and report any rash or mucosal lesions.
- Monitor INR if on warfarin.

PATIENT EDUCATION

- Drink plenty of water.
- Stop and seek help for rash, blisters, mouth ulcers, fever.
- Report sore throat, bruising or yellow skin.
- Avoid potassium supplements unless prescribed.

KEY POINTS

- TMP-SMX: folate pathway double blockade.
- Hyperkalaemia, SJS, marrow suppression.
- Drug of choice for Pneumocystis.
- Dangerous with methotrexate.